Search Algorithms

Warehouse Delivery Robot | Team 1- 3 members

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1. Task Problem

A robot in a warehouse must move from the start cell S to the goal cell G on a grid map. It moves Up, Down, Left, or Right (one cell at a time) and cannot enter a wall. Normal floor costs 1 and rough floor costs 3 for each move into that cell. Find a path from S to G using search algorithms.

Symbol	Meaning
S / G	Start cell / Goal cell (cost of entering G is 1; S is not counted)
. ~ #	Normal floor (cost 1) / Rough floor (cost 3) / Wall (cannot enter)

Maps are given as a variable at the top of your program (list of strings, one string per row).
Copy these example maps and use them for testing:

```
MAP_A = [
  "S..#....",
  "##.#.##.",
  "....~~..",
  ".####.#.",
  "...~..#G"
]

MAP_B = [
  "S~~~~~~~G",
  ".#######.",
  ".........",
  ".#.#.#.#.",
  "........."
]

MAP_C = [
  "S..#....",
  ".#.#.##.",
  "...#..#.",
  ".###.##.",
  "....#..G"
]
```

The map is made of cells. S is where the robot starts and G is where it must go. A dot . is normal floor (cost 1), a tilde ~ is rough floor (cost 3), and a hash # is a wall that the robot cannot pass through.

MAP_C has no solution, and MAP_B contains a trap: the shortest path is not the cheapest.

2. Requirements

- Implement at least TWO search algorithms: one uninformed (BFS, DFS, or UCS) and one informed (Greedy Best-First or A*). Do not use path-finding libraries.
- Define the map as a variable at the top of the program (e.g., GRID = MAP_A). Find S and G by scanning the grid.
- For each algorithm, print: the algorithm name, the path as a list of (row, column) from S to G, the total path cost, and the number of nodes expanded. If no path exists, print "No path found".
- Display the path on the map (e.g., mark path cells with *).
- For the informed algorithm, use a heuristic (e.g., Manhattan distance).
- Run every algorithm on MAP_A, MAP_B, and MAP_C. Record path length, path cost, nodes expanded, maximum frontier size, and run time in a table.
- Compare the two algorithms in a short paragraph or table: which one found the cheaper path, which one expanded fewer nodes, and which one you would choose for this robot and why.

3. Deliverables

#	Deliverable	Description
1	Source code	Code + Commented code, with instructions on how to run it and how to change the map variable.
2	Report	Problem formulation, algorithm descriptions, heuristic, results table, path screenshots, comparison, conclusion and github link.